Universitatea POLITEHNICA din București
Facultatea de Automatică și Calculatoare

Electronică analogică

Curs – Modulul 11

Filtre – introducere

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2022-2023

Sumar

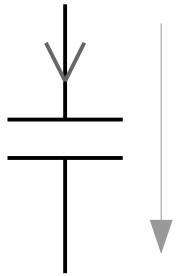
- Descriere comportamentală
  - domeniu timp, domeniu frecvență
  - clasificare
- Aplicații
  - audio
  - radio
- Descriere matematică
  - filtre pasive RC, RLC

Definiție

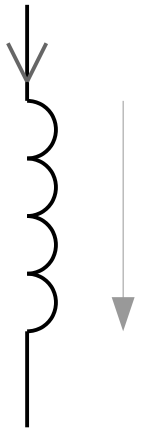
- Subcircuit cu comportament:
  - independent de amplitudinea semnalului (**linear**)
    - într-un domeniu de intrare specificat
  - dependent de viteza de variație a semnalului
    - **în mod intenționat**
    - toate dispozitivele reale manifestă e.g. capacități parazite

Exemple de dispozitive

având comportament dependent de **viteza de variație** a semnalului

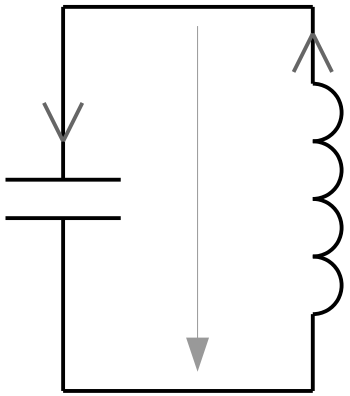


$$i_C = C\frac{dv_C}{dt}$$



$$v_L = L\frac{di_L}{dt}$$

Circuit rezonant (1)



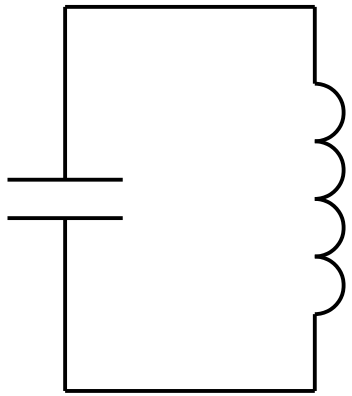
$$v = -LC\frac{d^2 v}{dt^2}$$

[timp^2]

$$\omega = \frac{1}{\sqrt{LC}}$$

$$v = -\frac{1}{\omega^2}\frac{d^2 v}{dt^2}$$

Circuit rezonant (2)



$$v = -\frac{1}{\omega^2}\frac{d^2 v}{dt^2}$$

Soluție:

$$v = V_0 \cos(\omega t + \varphi)$$

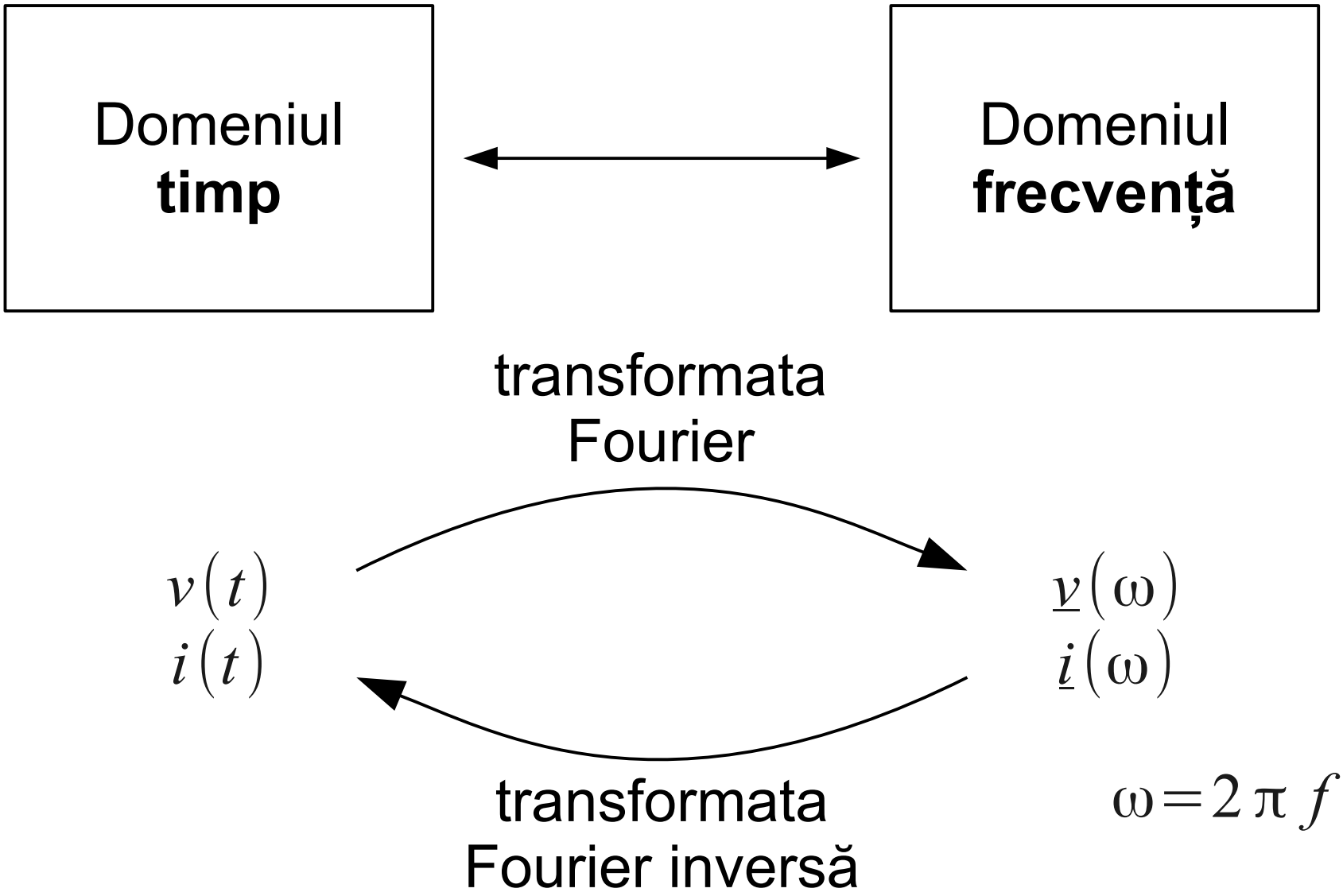
$$\frac{dv}{dt} = -\omega V_0 \sin(\omega t + \varphi)$$

$$\frac{d^2 v}{dt^2} = -\omega^2 V_0 \cos(\omega t + \varphi) = -\omega^2 v$$

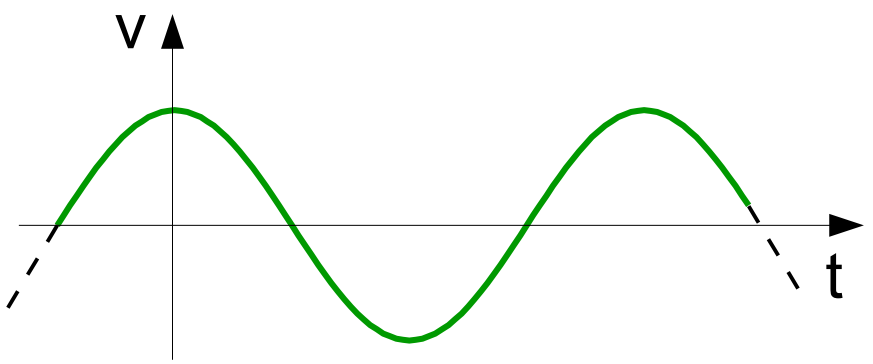
Comentarii

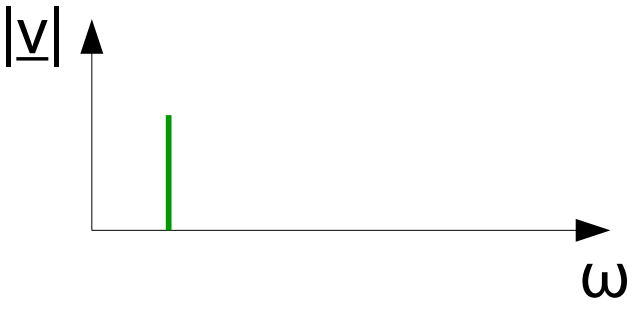
- Analogie mecanică: ansamblu masă – resort
  - energie cinetică <=> energie magnetică
  - energie p. gravitațională <=> energie p. electrică
- În realitate
  - forțe de frecare / rezistență electrică serie
  - oscilație amortizată
- Sinusoida – semnal “natural”
  - Fourier:
    - orice semnal = sumă (infinită) de sinusoide
      având diverse frecvențe, amplitudini și faze inițiale

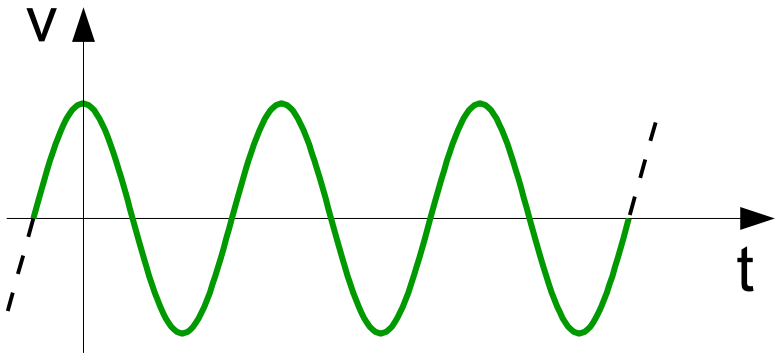
Domeniul frecvență

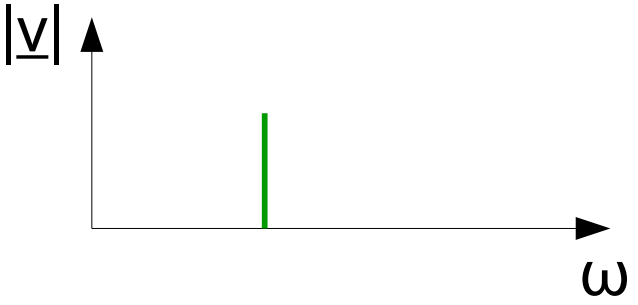


Spectrul de frecvență (1)

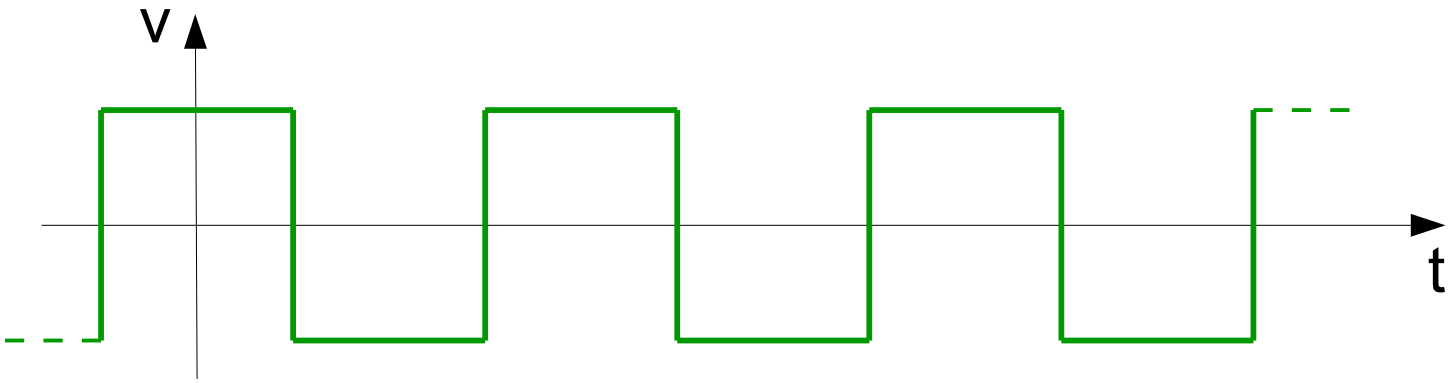


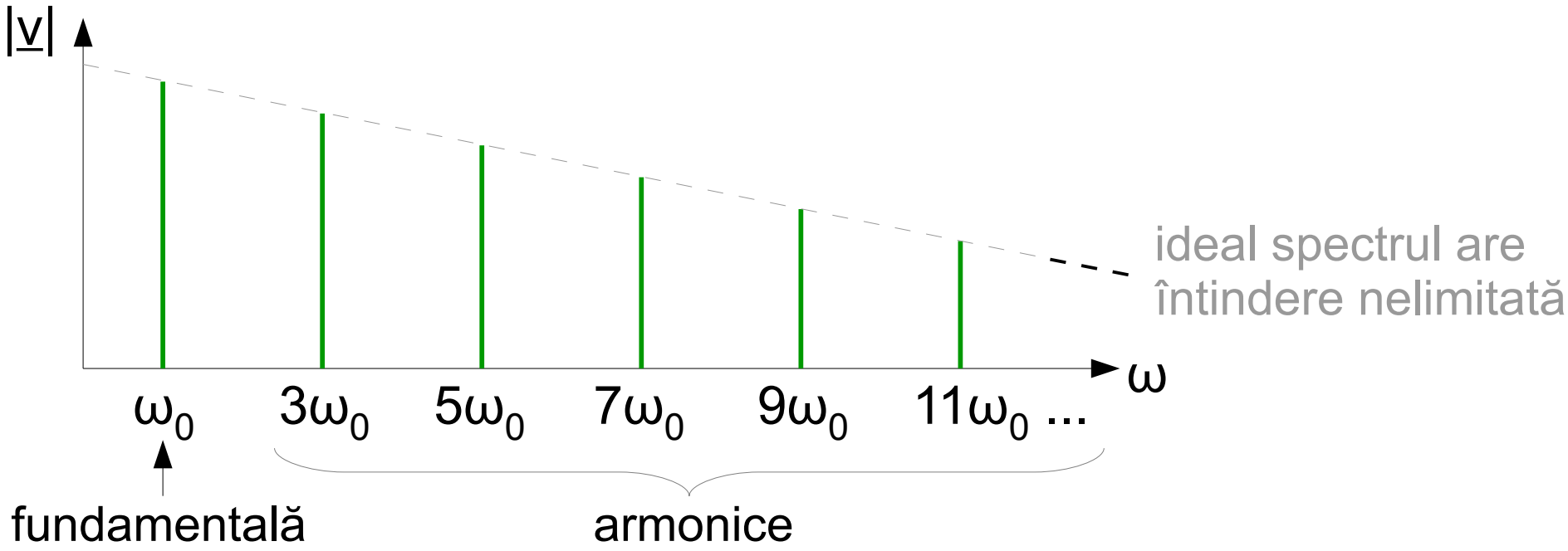




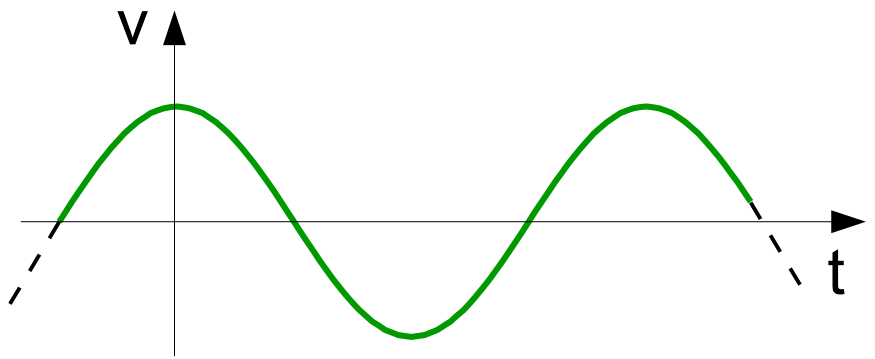


Spectrul de frecvență (2)

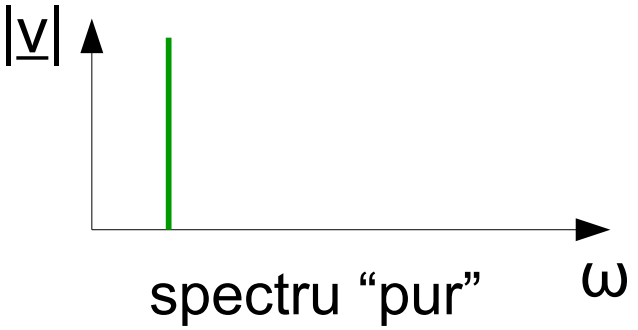




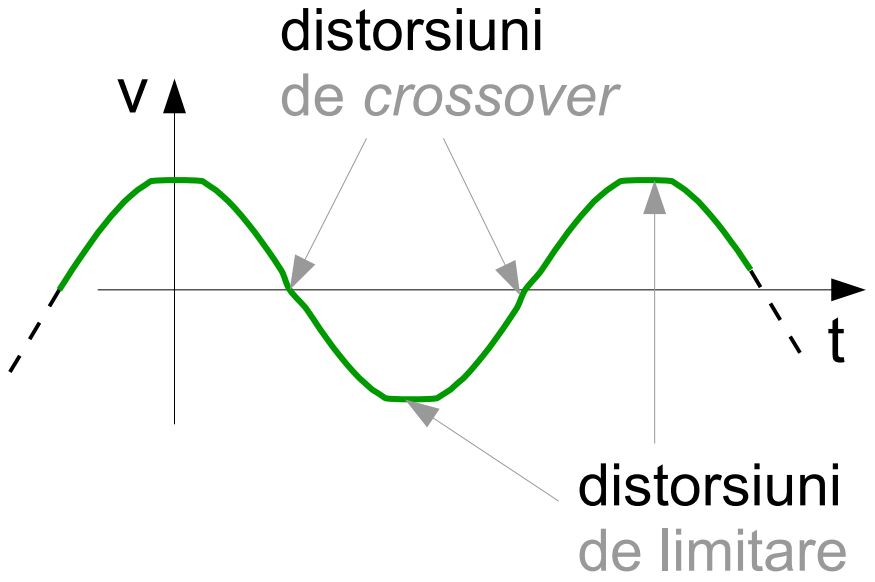
Spectrul de frecvență (3)

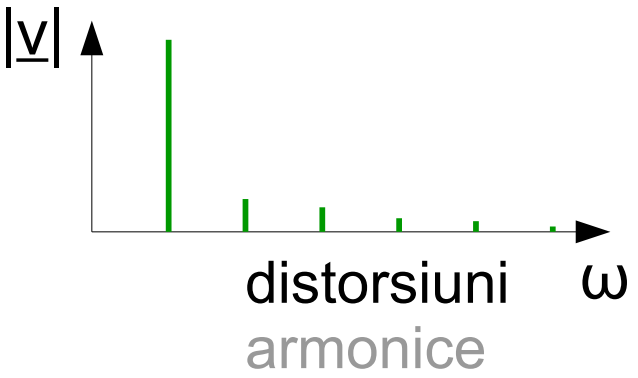


sinusoidă “perfectă”



spectru “pur”

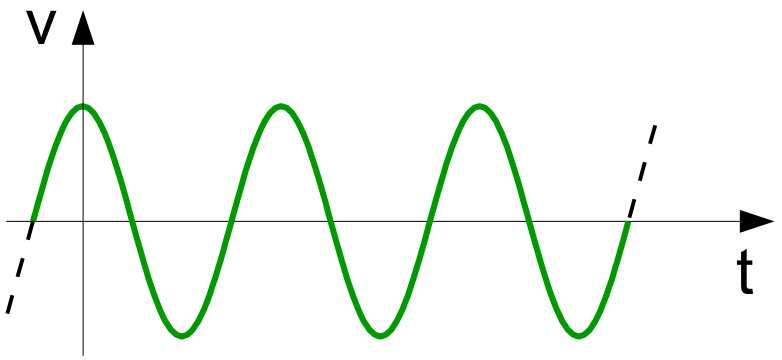




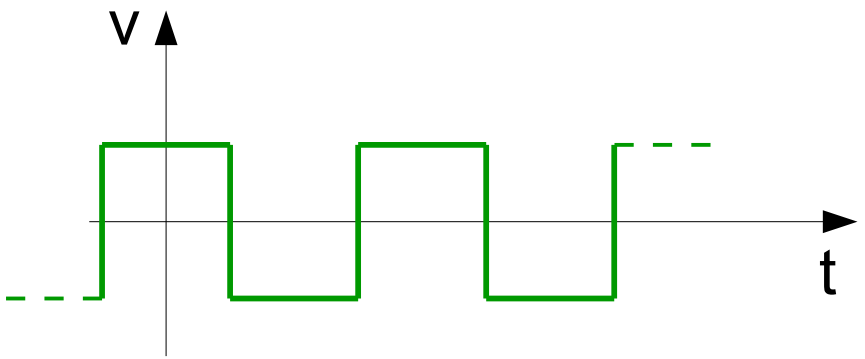
Specificație pentru amplificatoare:
Total Harmonic Distortion (THD)

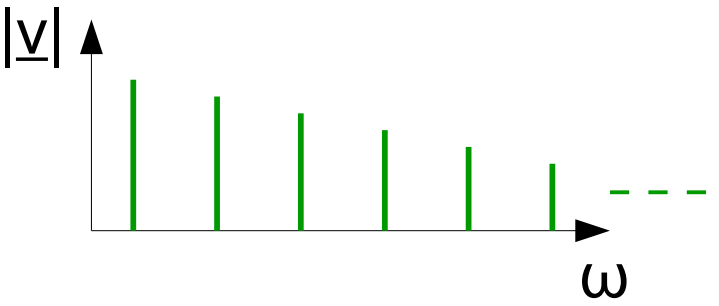
Spectrul de frecvență (4)

- Semnalele **periodice** au **spectru discret**



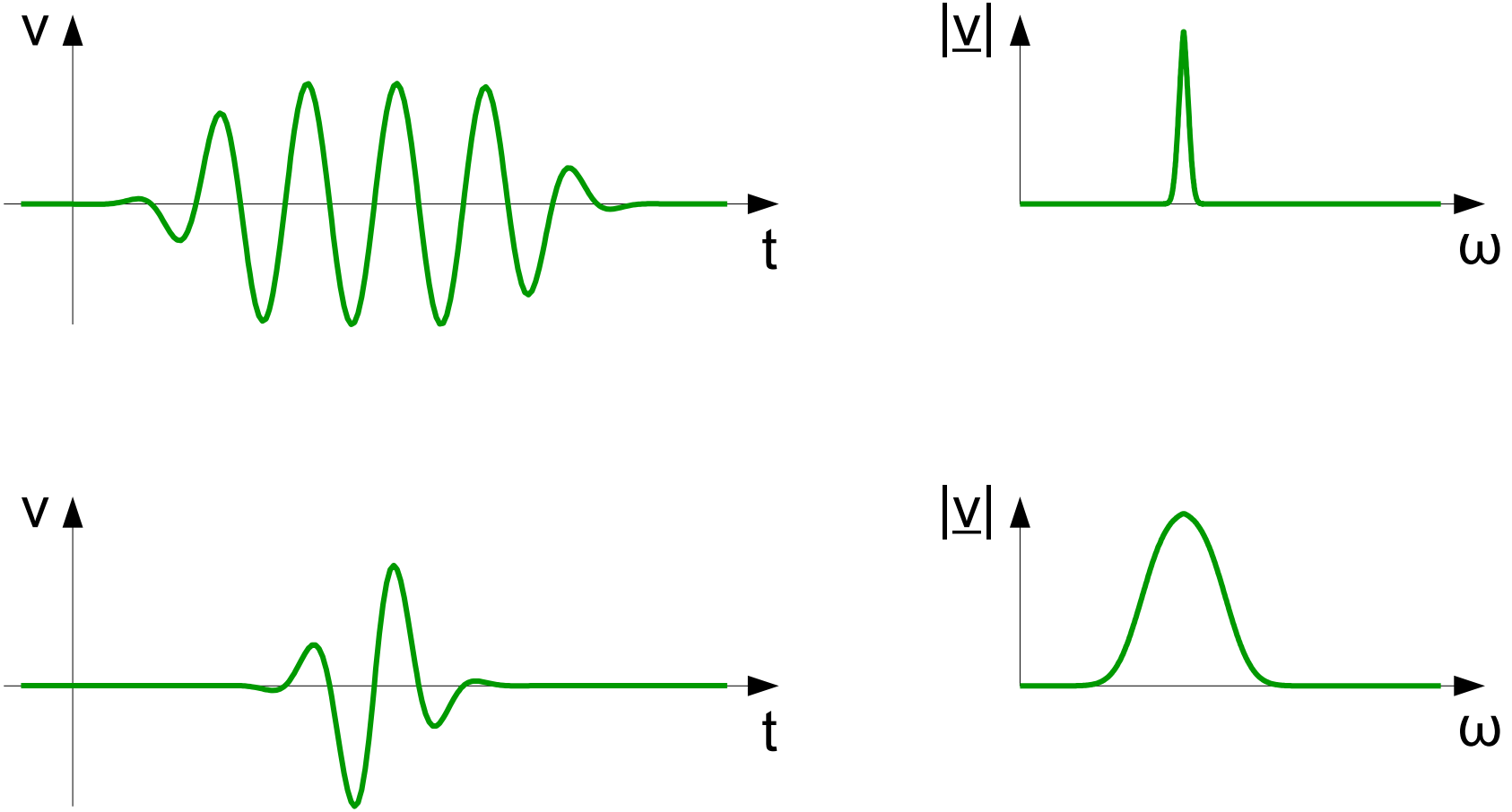






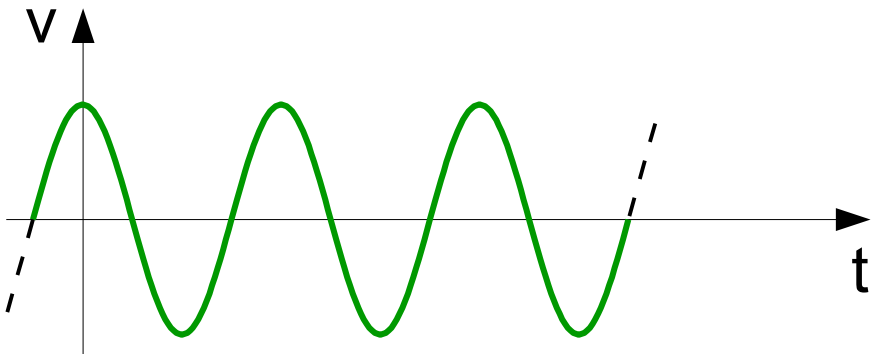
Spectrul de frecvență (5)

- Semnalele **aperiodice** au **spectru continuu**



Spectrul de frecvență (6)

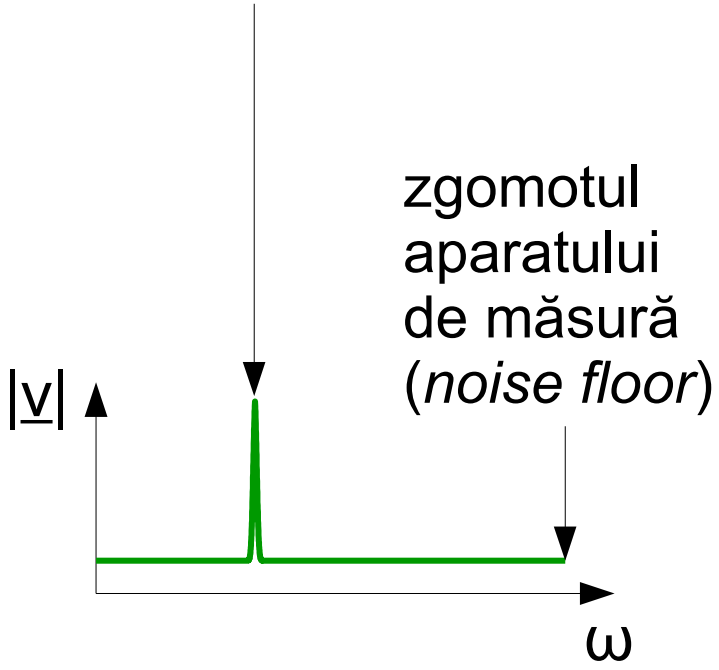
- în practică



Analizor de spectru

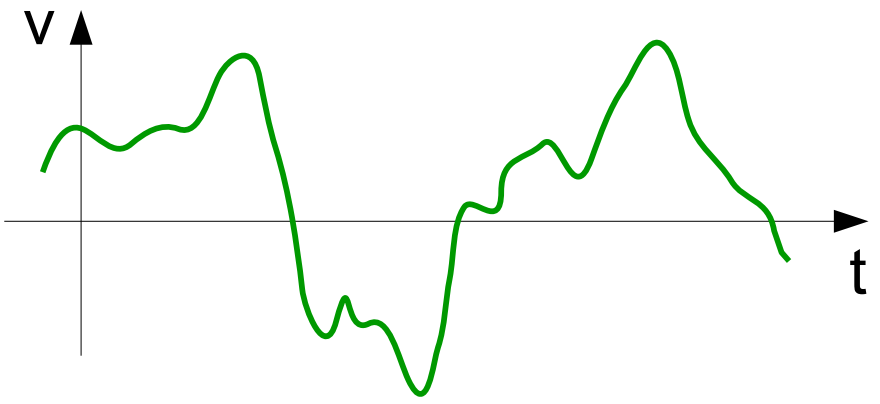
lățimea liniei spectrale (*peak*-ului) e dată de:

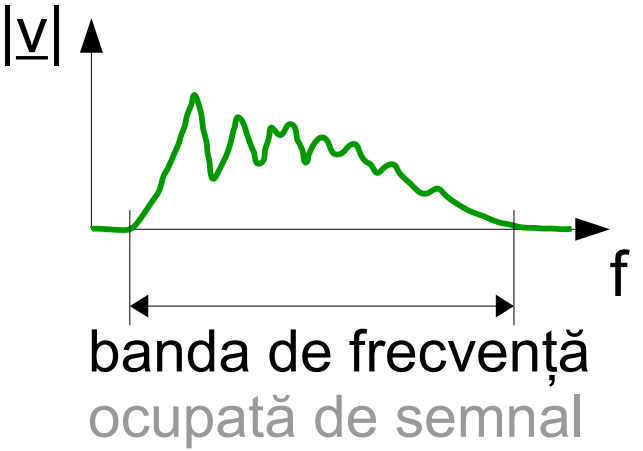
- rezoluția aparatului de măsură
- stabilitatea frecvenței semnalului studiat



Spectrul de frecvență (7)

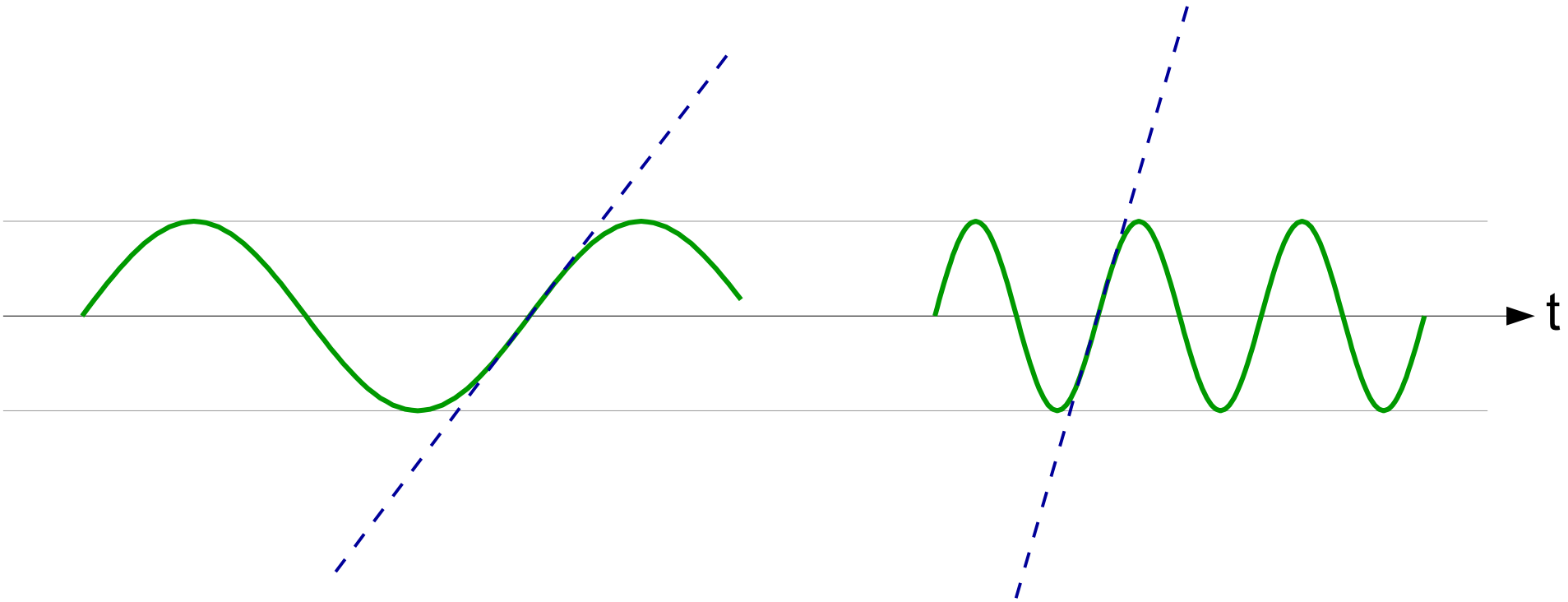
- Semnalele **aperiodice** au **spectru continuu**
  - semnalele “complicate” au spectru “complicat”
  - exemplu: semnale audio reale
  - circuite **lineare** => componentele din spectru sunt prelucrate independent (nu se influențează între ele) => este suficientă analiza din perspectiva unei singure componente sinusoidale (cu parametru ω)





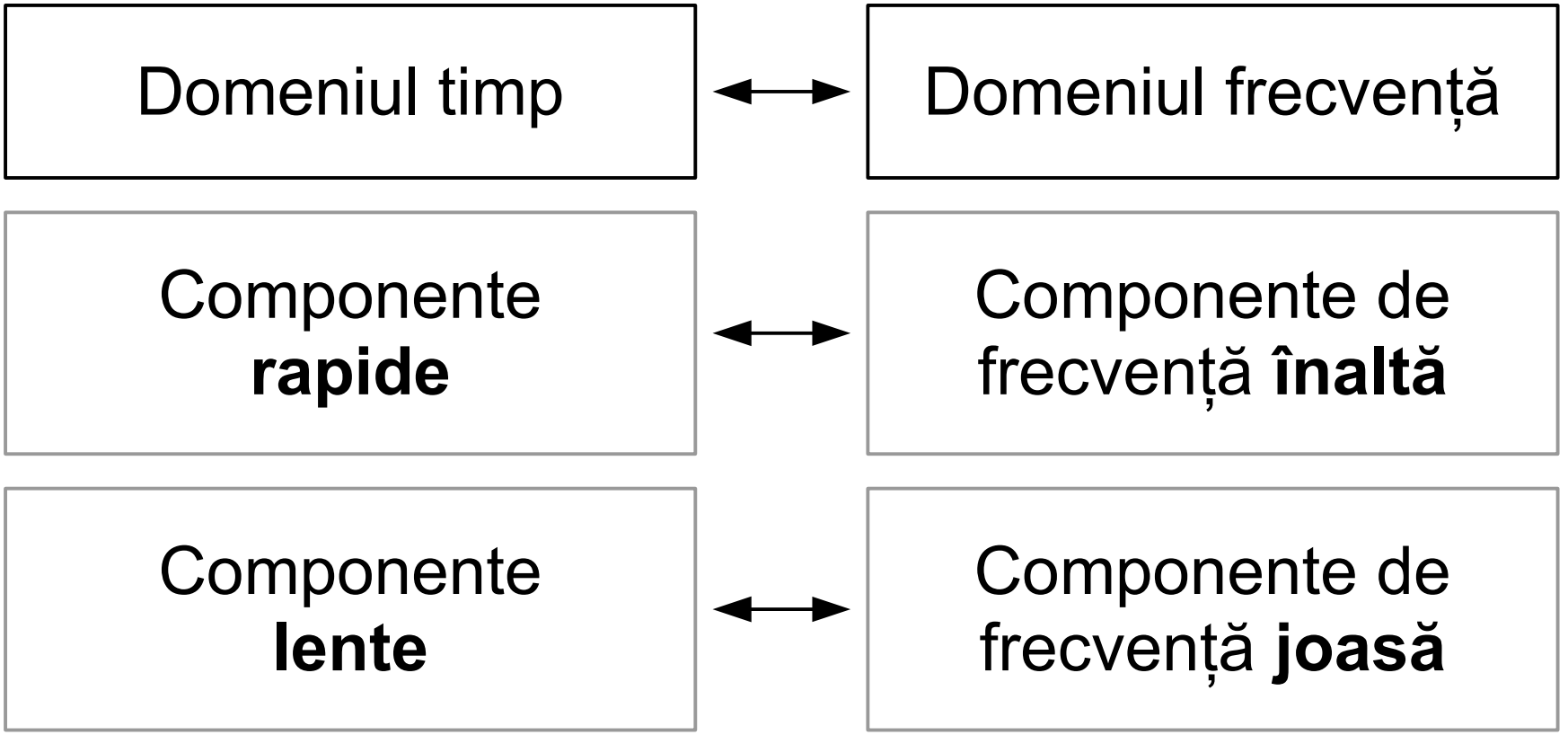
Frecvență <=> viteză de variație (1)

- relația se observă pe semnale sinusoidale având aceeași amplitudine

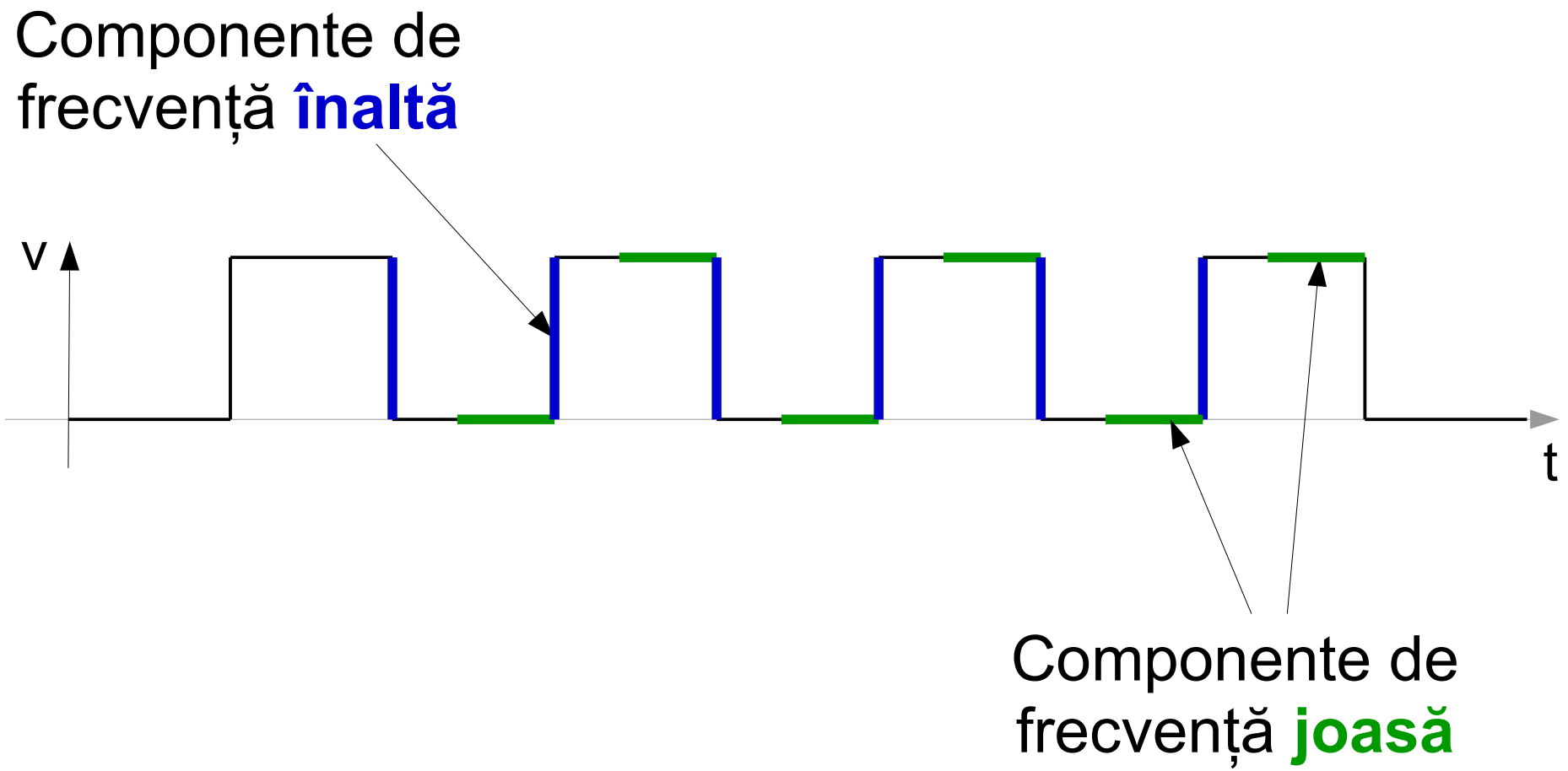


Frecvență <=> viteză de variație (2)

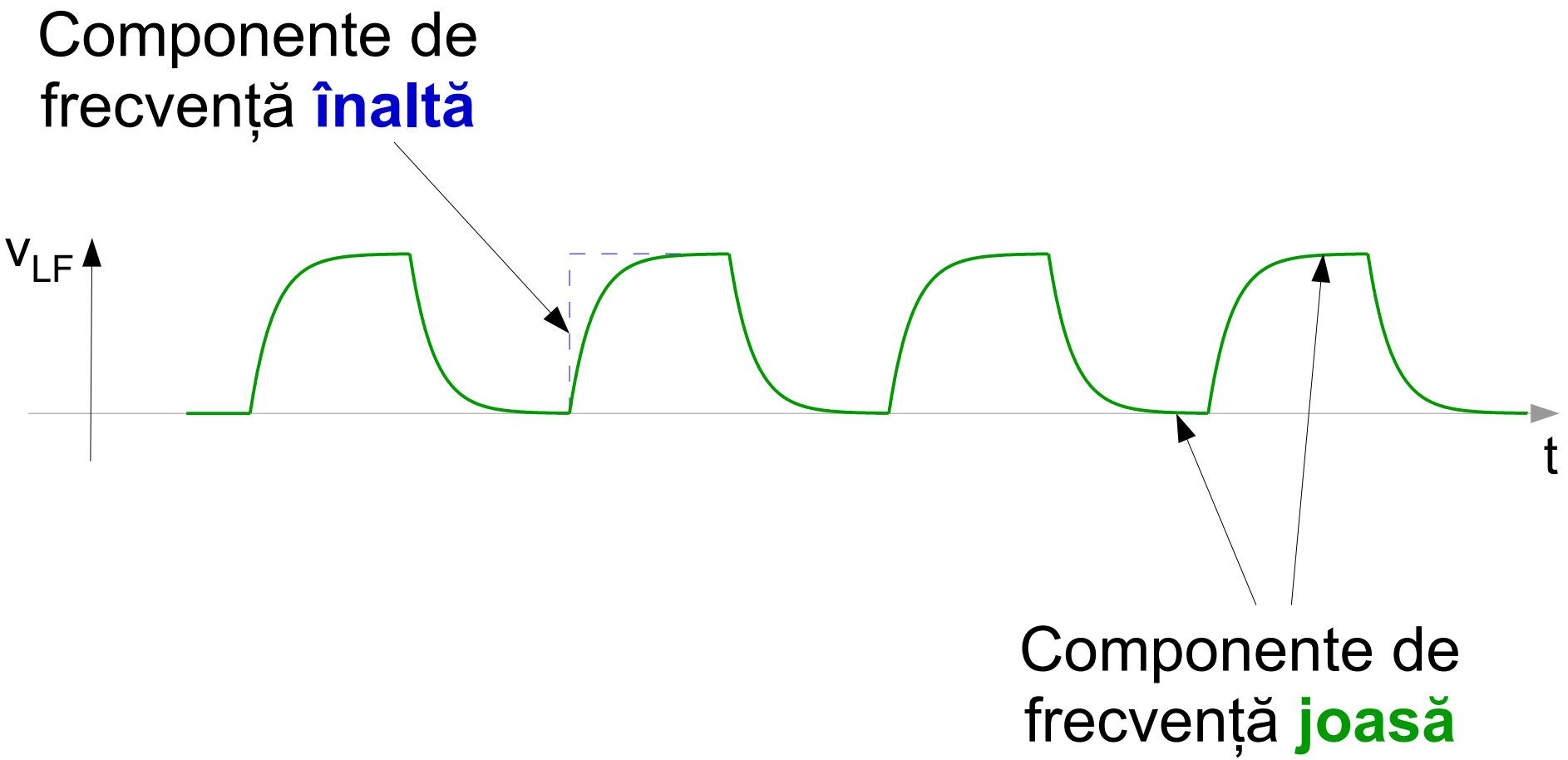
- și se generalizează (grație descompunerii Fourier) pentru **orice** formă a semnalului



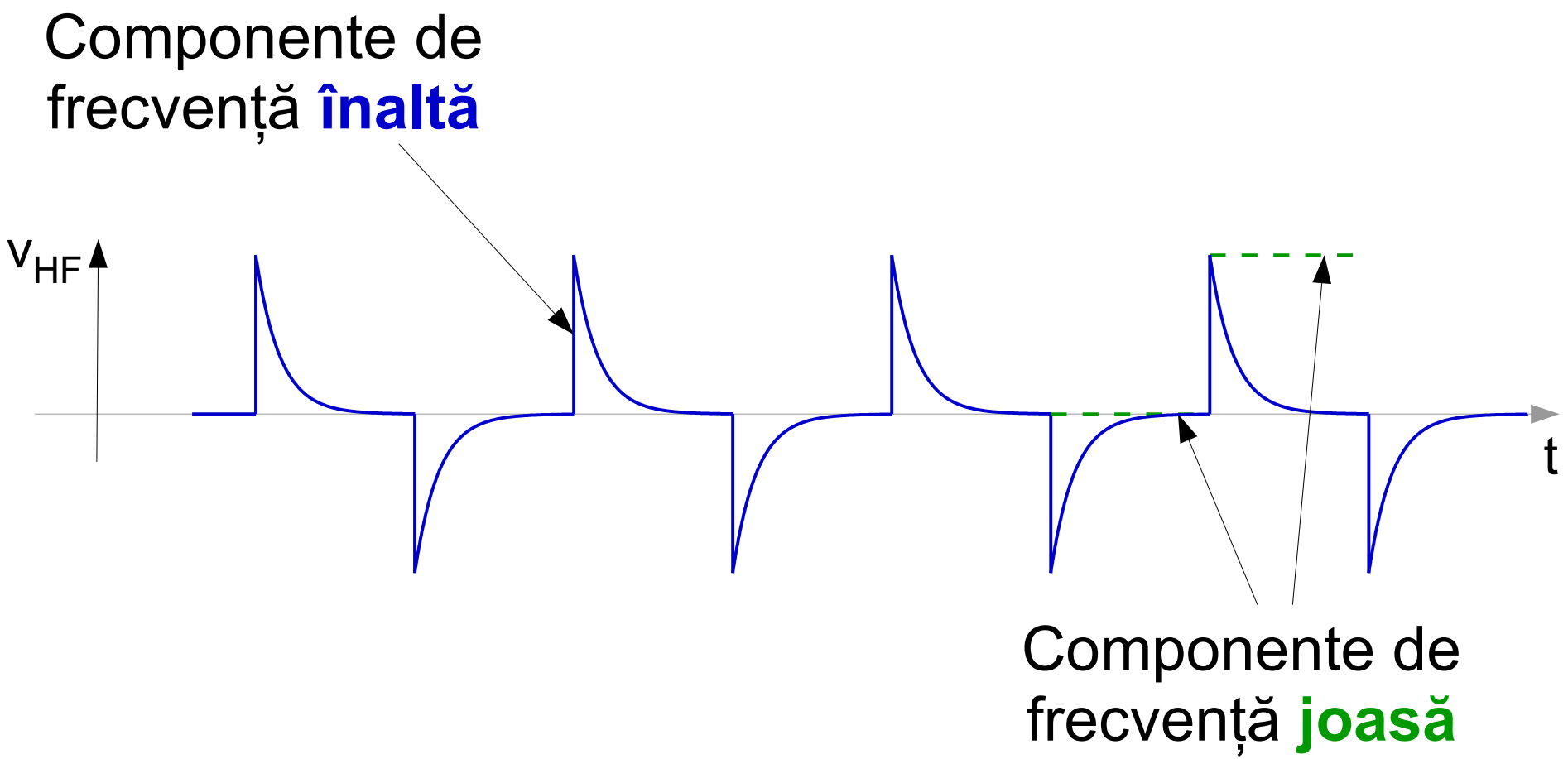
Ilustrare: semnal dreptunghiular



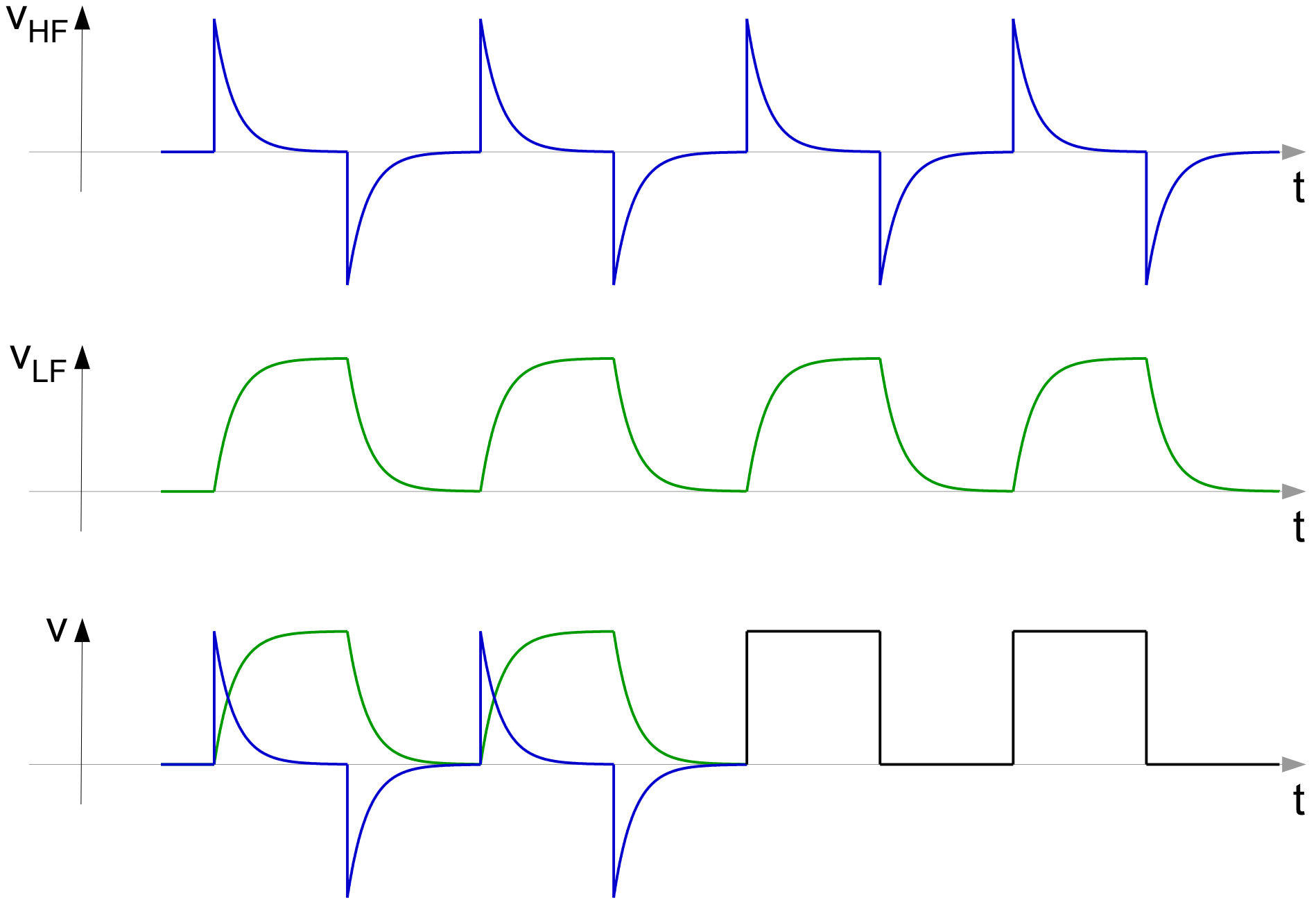
Filtrare trece-jos



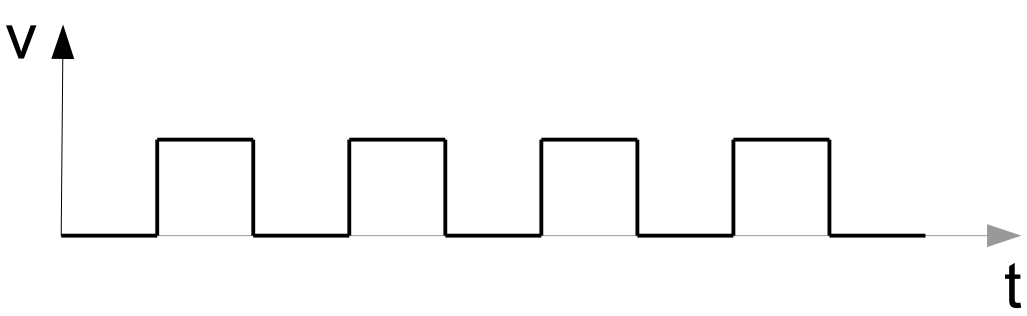
Filtrare trece-sus

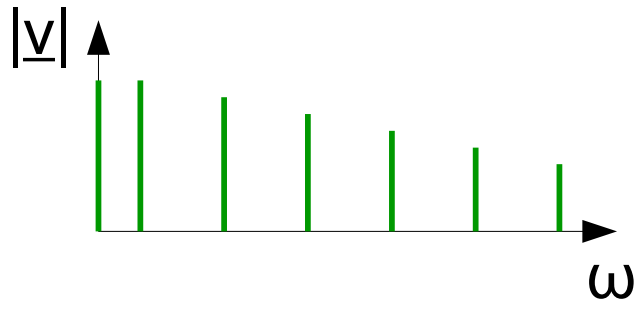


Componente rapide + lente

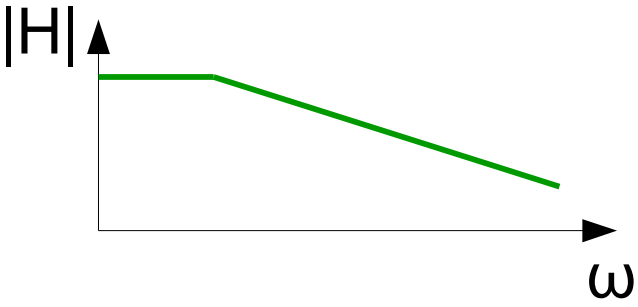


FTJ – domeniu frecvență



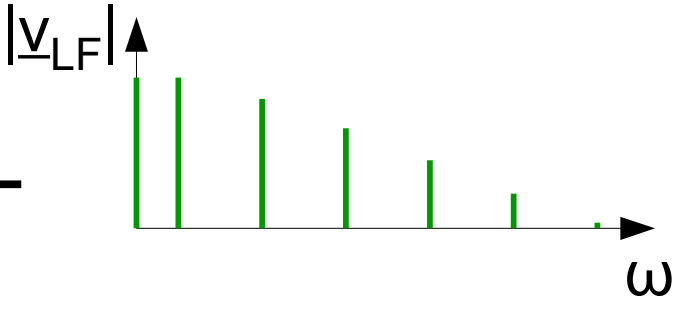




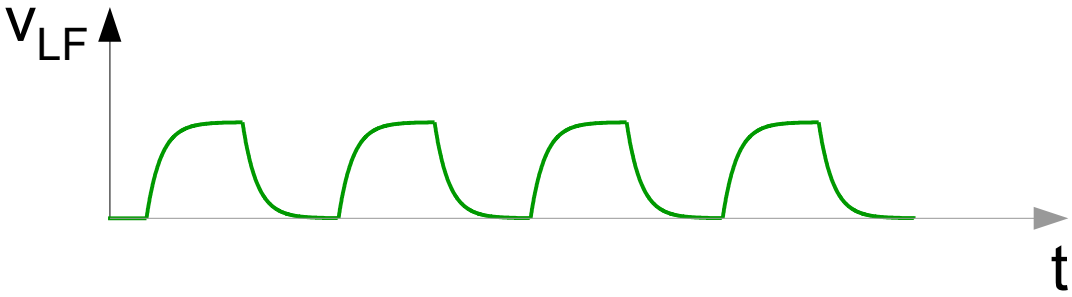


Funcția de transfer a filtrului trece-jos =>
FTJ modifică spectrul semnalului



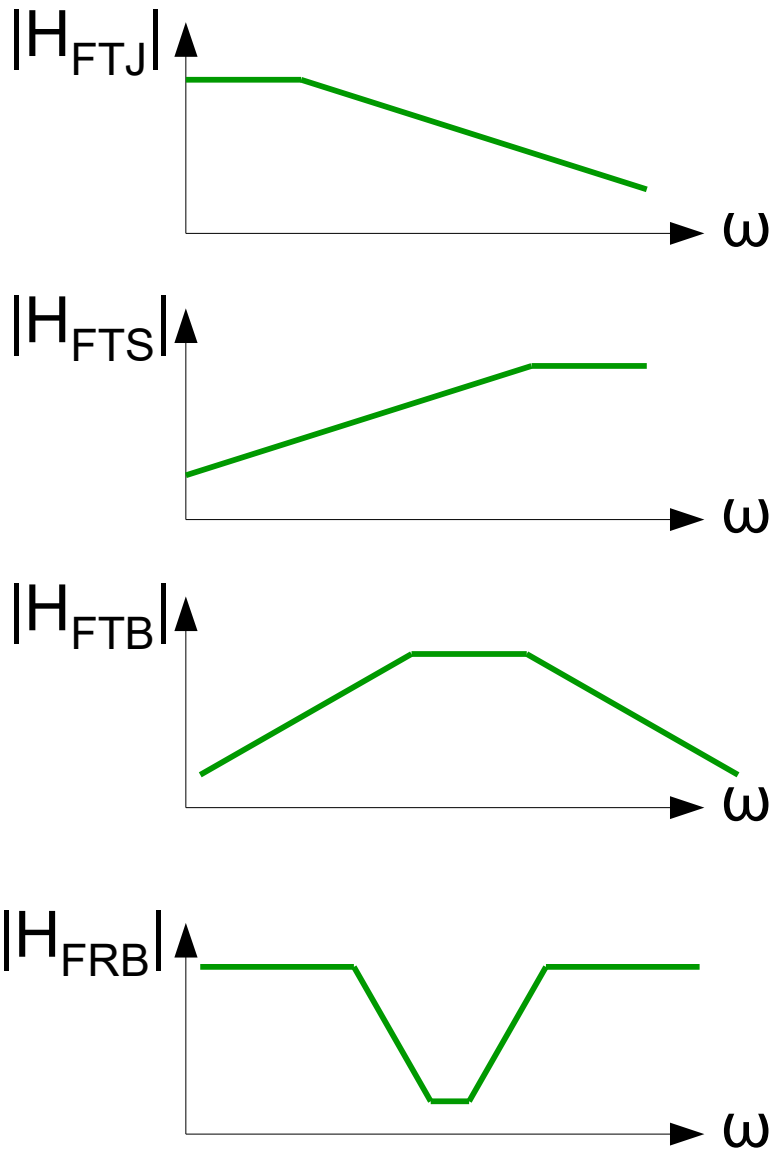






Clasificare

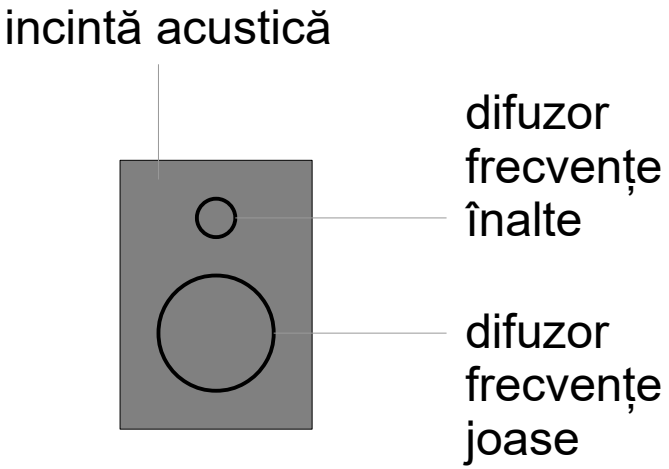
- Filtre
  - trece-jos
  - trece-sus
  - trece-bandă
  - rejecție-bandă
  - trece-tot (defazaj dependent de frecvență)
  - etc.

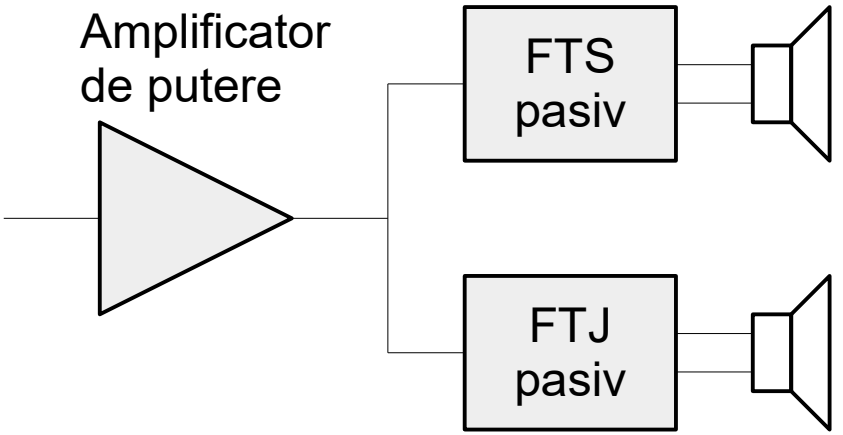


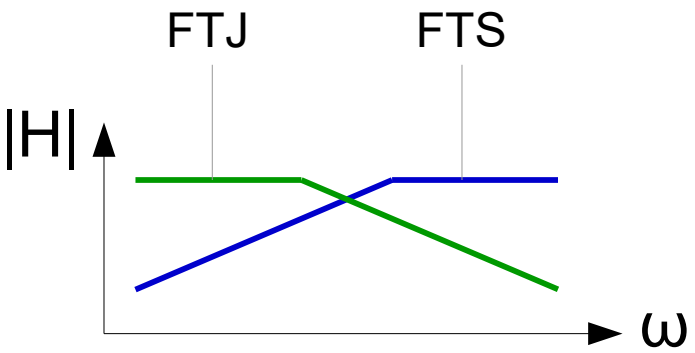
Aplicație – *crossover* audio (1)

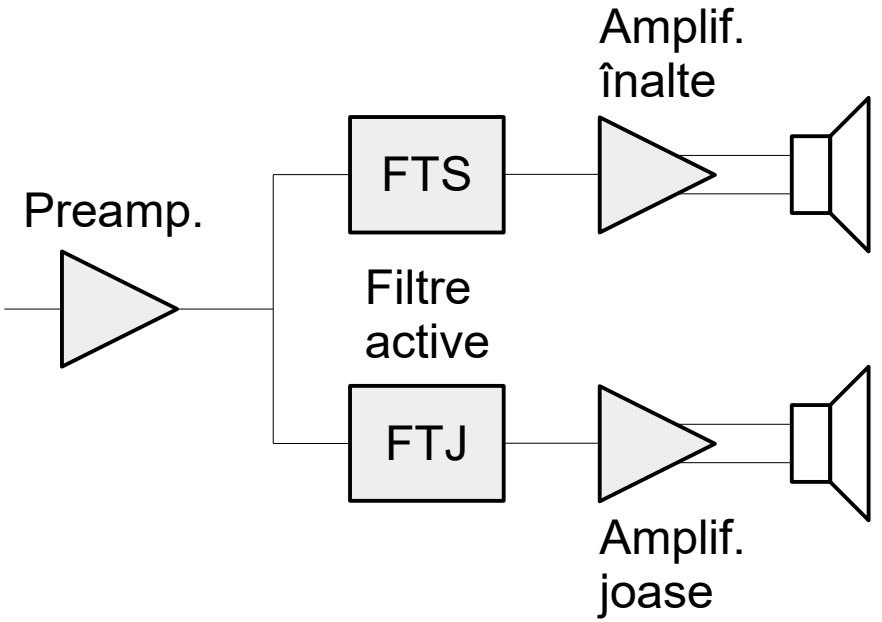
- Difuzor
  - reproduce sunetul analog cu semnalul electric aplicat, ideal în mod independent de frecvență
- în practică
  - comportament dependent de frecvență
  - aproximativ constant într-o bandă restrânsă
- de obicei
  - difuzor mare – frecvențe mici
  - difuzor mic – frecvențe mari

Aplicație – *crossover* audio (2)

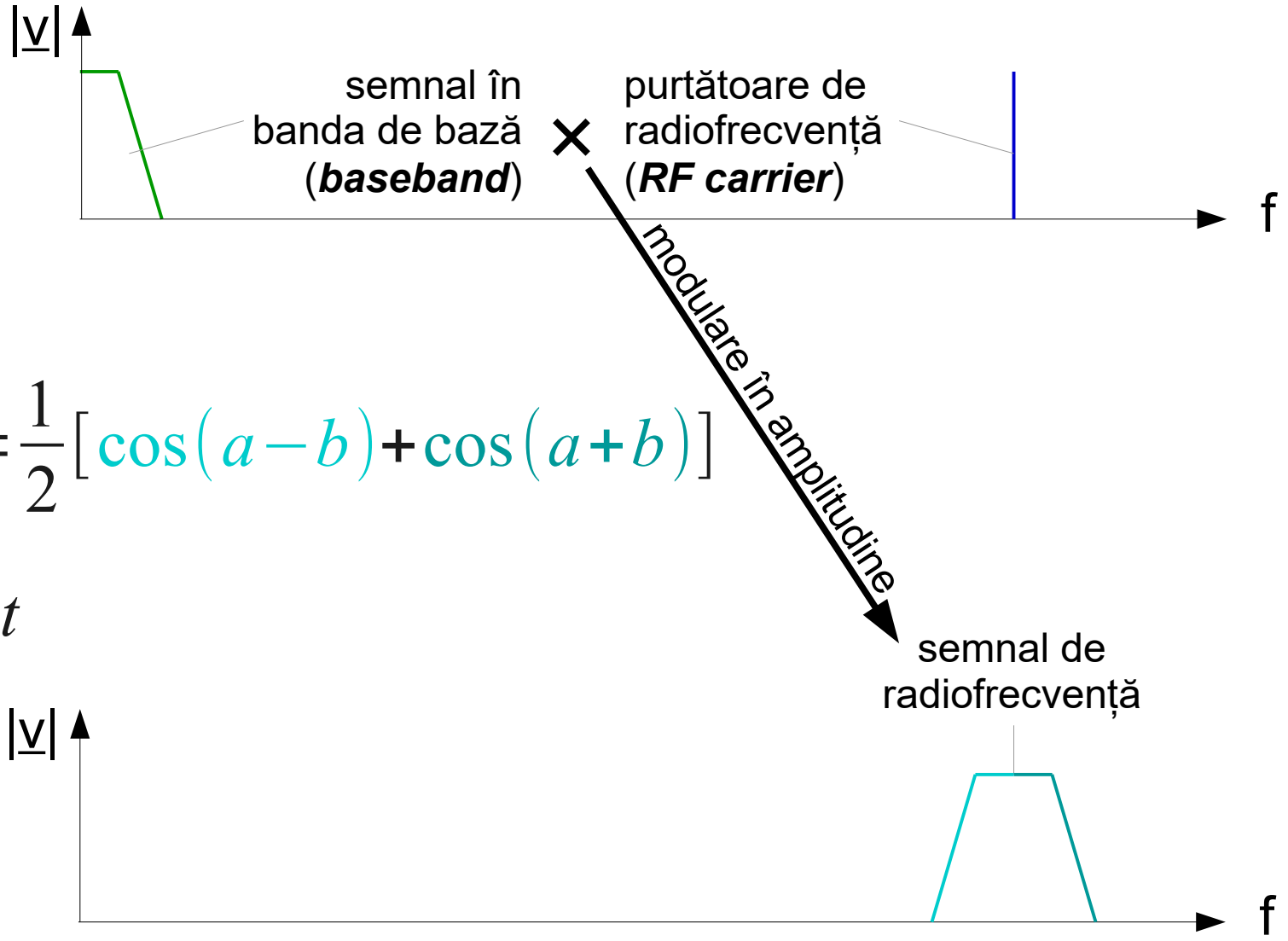








Aplicație – comunicația radio (1)

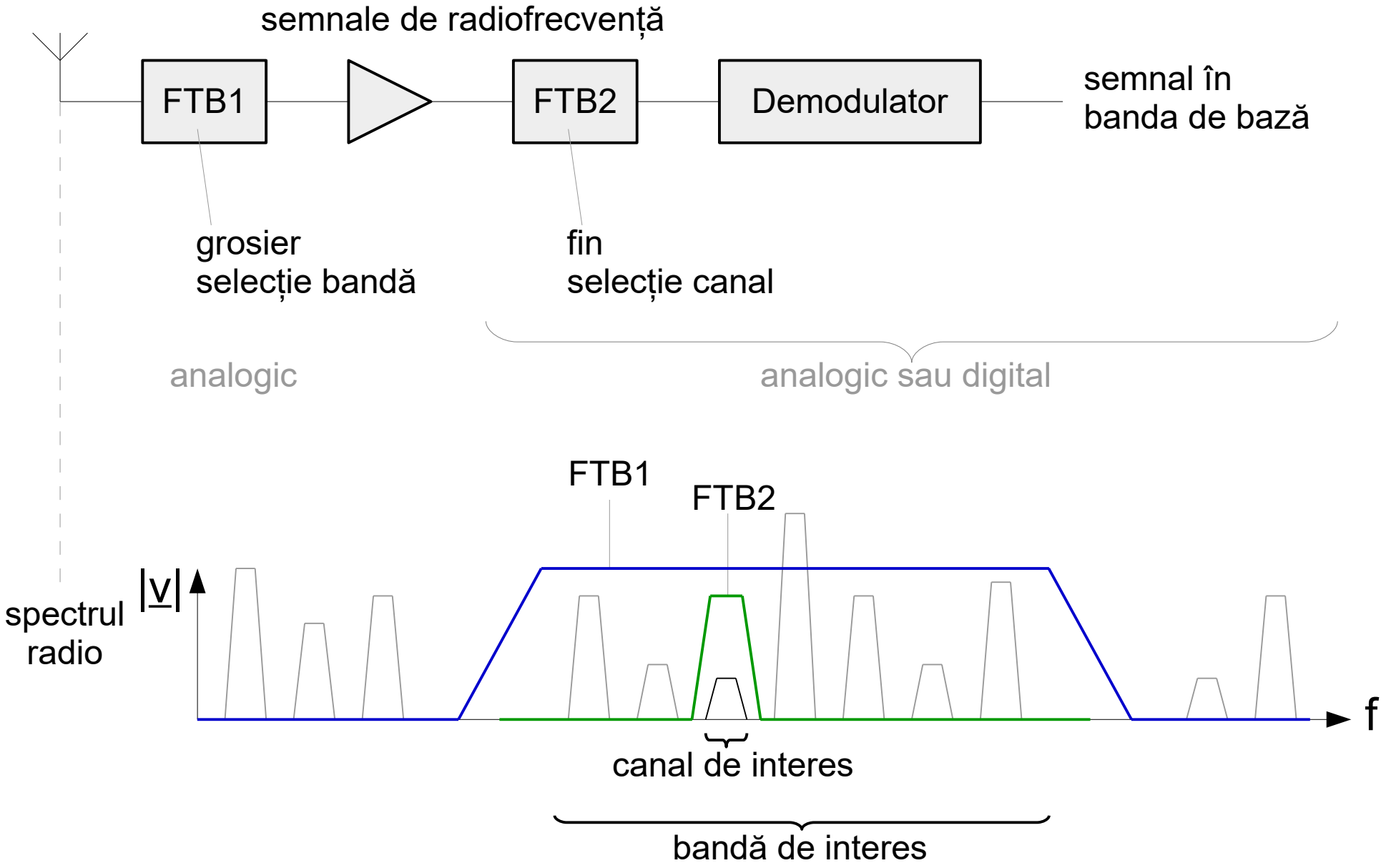


$$\cos a \cos b = \frac{1}{2}[\cos(a-b)+\cos(a+b)]$$

$\omega_{baseband}\, t$ $\omega_{RF}\, t$

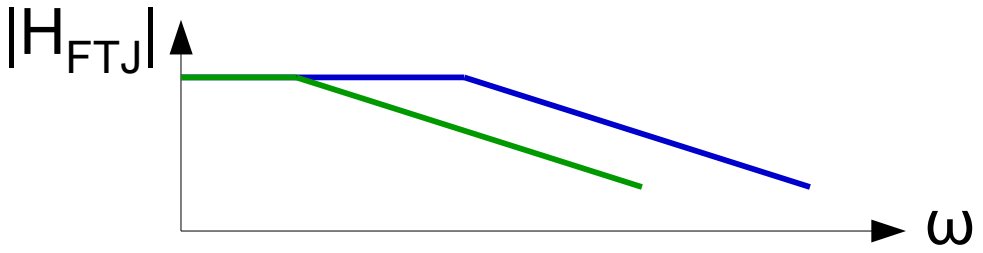
pentru fiecare componentă a semnalului

Aplicație – comunicația radio (2)

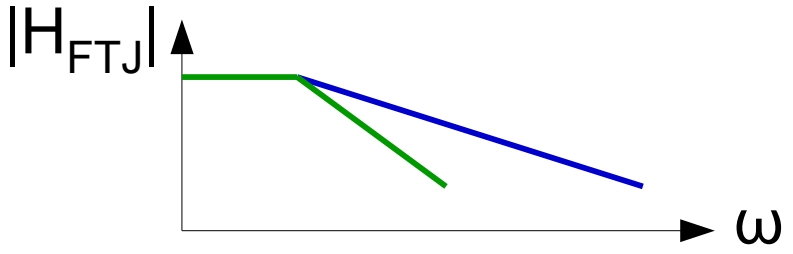


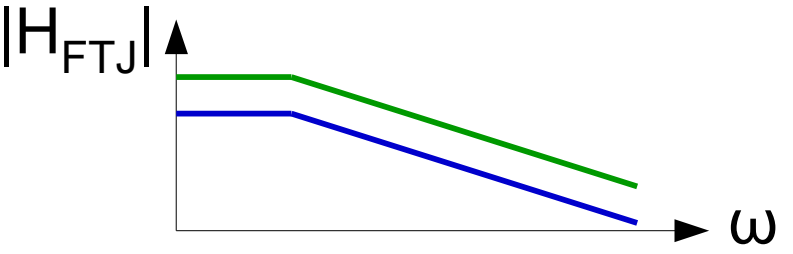
Parametrii principali

- Banda de trecere (sau de rejecție)



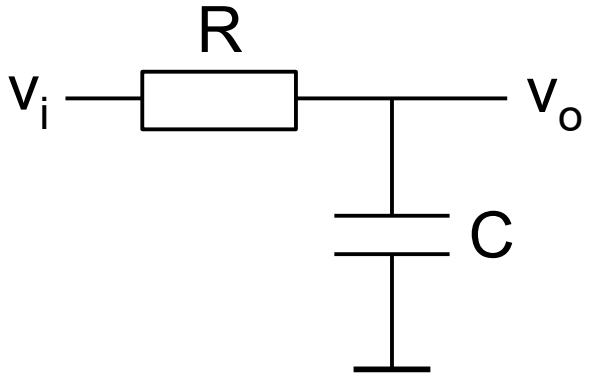
- Panta caracteristicii de frecvență
- Amplificarea în banda de trecere





FTJ pasiv RC

- Divizor de tensiune între impedanțe complexe



$$i_C = C\frac{dv_C}{dt} \rightarrow Z_C = \frac{1}{j\,\omega\,C}$$

$$H(\omega) = \frac{\underline{v_o}}{\underline{v_i}} = \frac{Z_C}{Z_C + R} = \frac{1}{1 + j\,\omega\,R\,C} = \frac{1}{1 + j\,\frac{\omega}{\omega_0}}$$

funcția de transfer

$$\omega_0 = \frac{1}{RC}$$

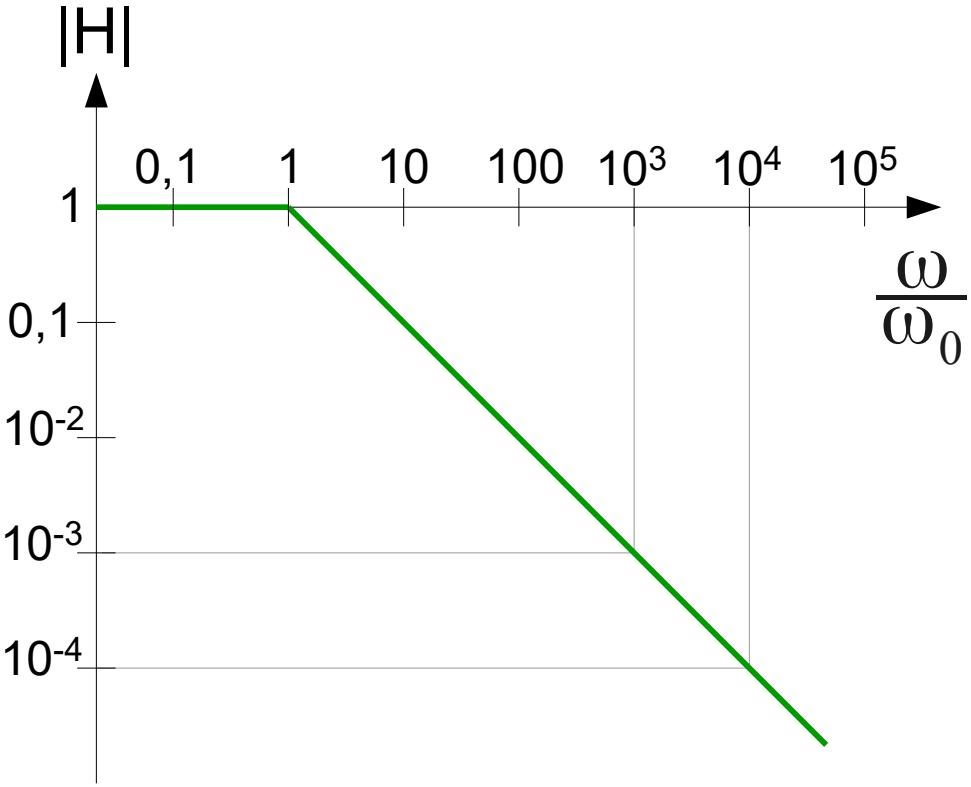
pulsație caracteristică

FTJ RC – caracteristica de frecv.

$$H(\omega)=\frac{1}{1+j\frac{\omega}{\omega_0}}=\frac{1-j\frac{\omega}{\omega_0}}{1+\frac{\omega^2}{\omega_0^2}} \qquad |H(\omega)|=\frac{1}{\sqrt{1+\frac{\omega^2}{\omega_0^2}}}$$

$$\omega \ll \omega_0 : \ |H(\omega)| \approx 1$$

$$\omega \gg \omega_0 : \ |H(\omega)| \approx \frac{\omega_0}{\omega}$$



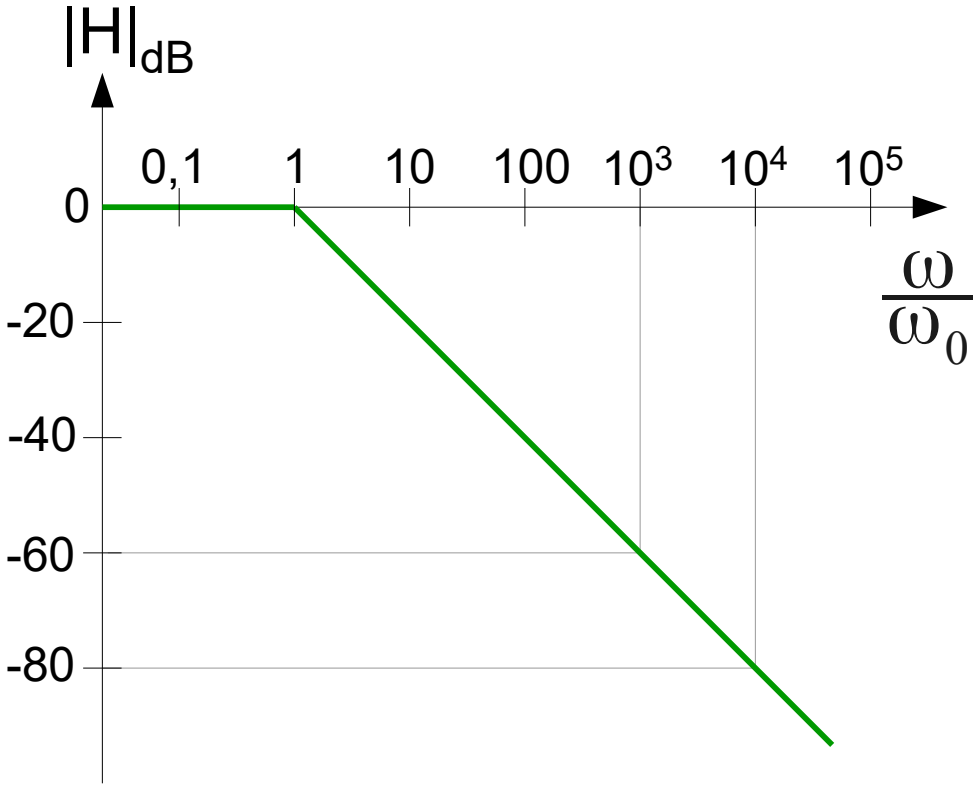
Exprimarea rapoartelor în dB

$$A_{p,dB} = 10 \log_{10} A_p$$ amplificare de putere

$$A_{v,dB} = 20 \log_{10} A_v$$ amplificare de tensiune

$$p = vi = \frac{v^2}{Z}$$

filtru de ordin I
panta –20 dB / decadă



Banda de trecere, punctul de -3 dB

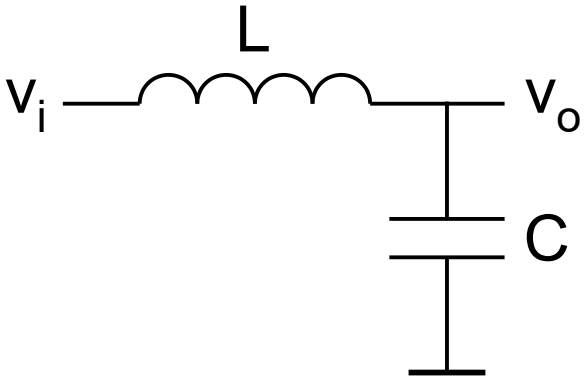
- În general banda de trecere a unui filtru se măsoară având ca repere punctele unde amplificarea de **putere** scade la jumătate
  - tensiunea scade de radical din 2 ori
  - se mai numesc puncte de -3 dB: $10\log_{10}\frac{1}{2}\approx -3$

- pentru FTJ de ordin I :

$$|H(\omega)|=\frac{1}{\sqrt{1+\frac{\omega^2}{\omega_0^2}}};\quad |H(\omega_{-3dB})|=\frac{1}{\sqrt{2}}|H(\omega\ll\omega_{-3dB})|$$

$$\omega_{-3dB}=\omega_0$$

FTJ pasiv LC



$$H(\omega)=\frac{Z_C}{Z_C+Z_L}=\frac{1}{1-\omega^2 LC}=\frac{1}{1-\frac{\omega^2}{\omega_0^2}}$$

funcția
de transfer

$$\omega_0=\frac{1}{\sqrt{LC}}$$

pulsația
caracteristică

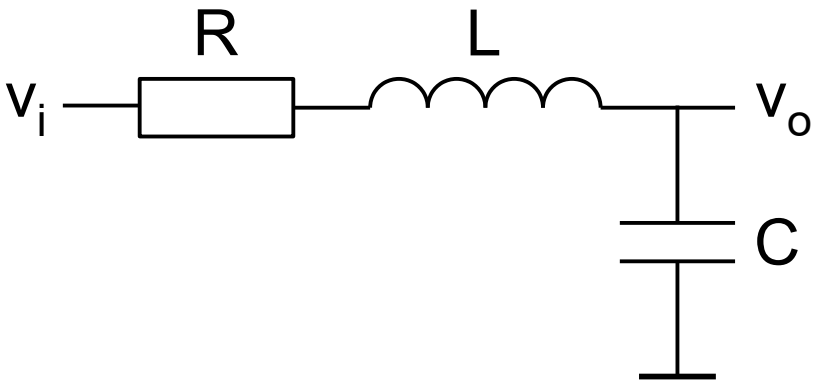
FTJ LC – caracteristica de frecv.

$$H(\omega)=\frac{1}{1-\frac{\omega^2}{\omega_0^2}}$$

$$\omega\to\omega_0 : \quad H\to\infty$$

frecvență de **rezonanță** – amplifică tensiunea “infinit” deoarece nu există elemente pe care să se disipe energia – se acumulează continuu în filtru (ideal)

FTJ pasiv RLC



$$H(\omega)=\frac{Z_C}{Z_C+Z_R+Z_L}=\frac{1}{1+j\,\omega\,RC-\omega^2\,LC}=\frac{1}{1+2\,j\,\zeta\,\frac{\omega}{\omega_0}-\frac{\omega^2}{\omega_0^2}}$$

$$\omega_0=\frac{1}{\sqrt{LC}}$$

$$\zeta=\frac{1}{2}\,RC\,\omega_0=\frac{R}{2}\sqrt{\frac{C}{L}}$$

factor de calitate

$$Q=\frac{1}{R}\sqrt{\frac{L}{C}}$$

FTJ RLC – caracteristica de frecv.

$$H(\omega)=\frac{1}{1+2\,j\,\zeta\,\frac{\omega}{\omega_0}-\frac{\omega^2}{\omega_0^2}}$$

$$|H(\omega_0)|=\frac{1}{2\,\zeta}$$

supracreștere (rezonanță)

$$H(\omega \ll \omega_0)\approx 1$$

$$H(\omega \gg \omega_0)\approx \frac{1}{2\,j\,\zeta\,\frac{\omega}{\omega_0}-\frac{\omega^2}{\omega_0^2}}\;;\quad |H(\omega \gg \omega_0)|\approx \frac{\omega_0^2}{\omega^2}$$

-40 dB / decadă